
01 Laboratory Exercise - The ScoreKeeper (Dynamic Array)

Student Name: Edgardo B. Del Rosario Jr.

Year and Section: BSIT-1C

Course Learning Objectives:

C01: Comprehend and clarify the characteristics, functions, and applications of essential data structures (such as arrays, linked lists, stacks, queues, trees, and graphs).

C02: Create and execute data structures and algorithms in a programming language.

C04: Utilize suitable data structures and algorithms to address computational challenges.

Activity Tasks:

- Create a program that handles a class size unknown at compile time, demonstrating the necessity of Heap memory allocation.

Submission Contents/ Requirements:

1. You must include the actual source code file and/or screenshots of the running program showing its successful operation.
2. Please ensure that your written answers or entries are provided on new, separate sheets.

Your PDF must be named as:

Class_LastName_FirstName_LE01.pdf (Ex: IT1a_DelaCruz_Juana_LE01.pdf)

Failure to follow instructions (wrong format, missing screenshot, no code inside PDF) will result in score deduction.

Scenario

You are writing a grading system for a professor. The class size changes every semester, so you cannot use a fixed-size array (e.g., `int scores[50]`). You must ask the user for the number of students, allocate the exact amount of memory needed, record their scores, and calculate statistics.

Requirements

1. Prompt the user for the number of students (**N**).
2. Dynamically allocate an integer array of size (**N**) using the **new** operator.
3. Input scores for all (**N**) students.
4. Calculate and print:
 - The **Highest** score.
 - The **Lowest** score.
 - The **Average** score.
5. **Critical:** Deallocate the memory using **delete[]** before the program exits to prevent memory leaks.

Sample Output

```
Enter class size: 5
Enter score #1: 85
Enter score #2: 92
Enter score #3: 78
Enter score #4: 90
Enter score #5: 88 --
- Report --- Highest
Score: 92 Lowest
Score: 78 Average
Score: 86.6
```

Code Skeleton

```
#include <iostream>
using namespace std;
int main() {

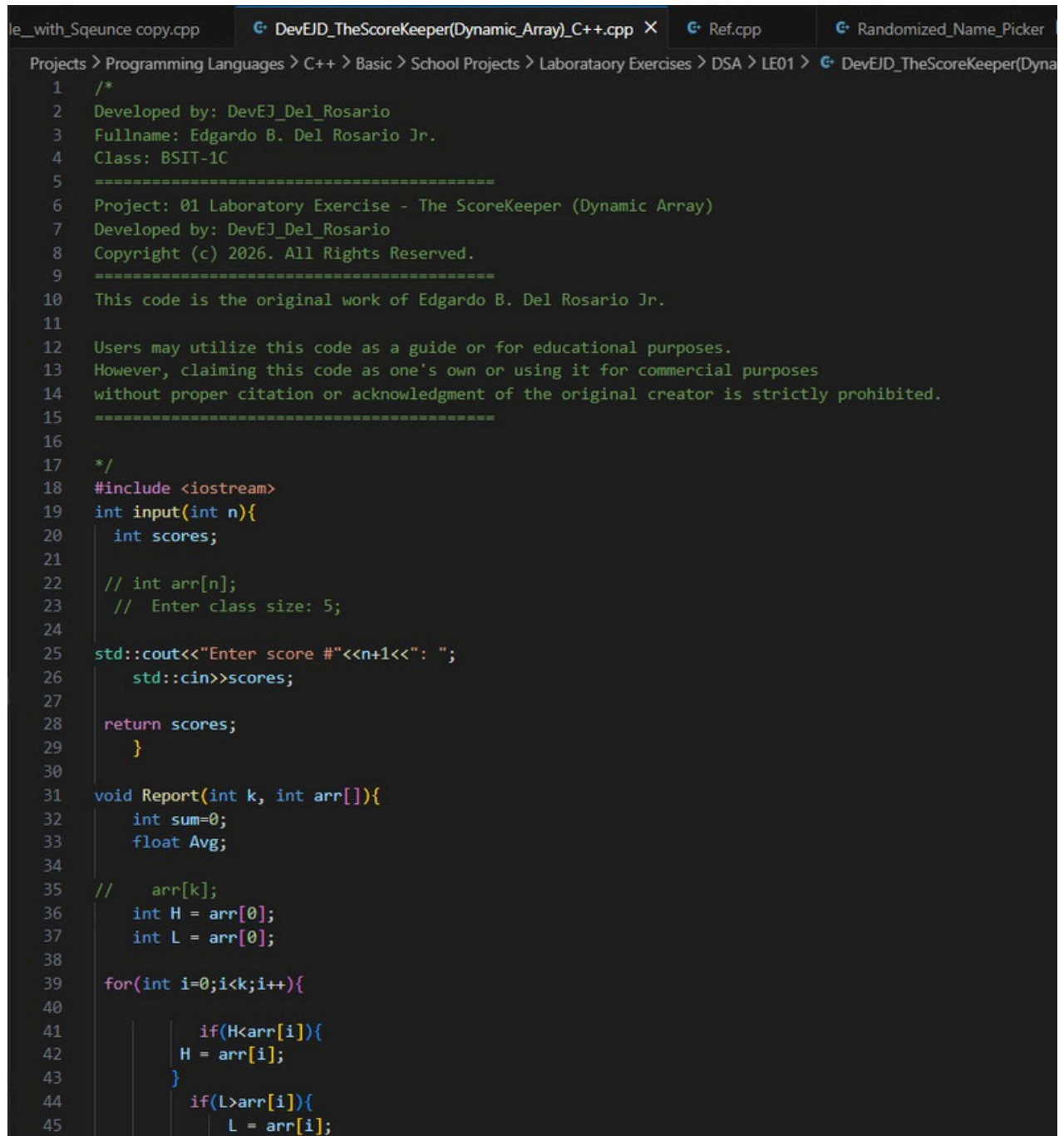
    int n;
    cout << "Enter class size: ";
    cin >> n;
    // TODO: Allocate memory
    // int* scores = ...
    // TODO: Input Loop
    // TODO: Calculate Stats
    // TODO: Deallocate memory
    return 0;

}
```

Criteria	Description	Points
Dynamic Allocation	Correct usage of <code>new int[n]</code> to allocate the array on the Heap based on user input.	15 pts
Data Processing	Loops correctly iterate through the array to input data and accurately calculate High, Low, and Average stats.	20 pts
Memory Safety	<code>delete[]</code> is explicitly called before program termination to prevent memory leaks.	15 pts
Total		50 pts

MY PROGRAM

SCREENSHOTS:



```
le_with_Squence copy.cpp  DevEJD_TheScoreKeeper(Dynamic_Array)_C++.cpp X  Ref.cpp  Randomized_Name_Picker

Projects > Programming Languages > C++ > Basic > School Projects > Laboratory Exercises > DSA > LE01 > DevEJD_TheScoreKeeper(Dyna

1  /*
2  Developed by: DevEJ_Del_Rosario
3  Fullname: Edgardo B. Del Rosario Jr.
4  Class: BSIT-1C
5  =====
6  Project: 01 Laboratory Exercise - The ScoreKeeper (Dynamic Array)
7  Developed by: DevEJ_Del_Rosario
8  Copyright (c) 2026. All Rights Reserved.
9  =====
10 This code is the original work of Edgardo B. Del Rosario Jr.
11
12 Users may utilize this code as a guide or for educational purposes.
13 However, claiming this code as one's own or using it for commercial purposes
14 without proper citation or acknowledgment of the original creator is strictly prohibited.
15 =====
16
17 */
18 #include <iostream>
19 int input(int n){
20     int scores;
21
22     // int arr[n];
23     // Enter class size: 5;
24
25     std::cout<<"Enter score #"<<n+1<<" : ";
26     std::cin>>scores;
27
28     return scores;
29 }
30
31 void Report(int k, int arr[]){
32     int sum=0;
33     float Avg;
34
35     // arr[k];
36     int H = arr[0];
37     int L = arr[0];
38
39     for(int i=0;i<k;i++){
40
41         if(H<arr[i]){
42             H = arr[i];
43         }
44         if(L>arr[i]){
45             L = arr[i];
```

```

46         }
47     }
48
49     std::cout<<"\n\nTotal Score:"<<std::endl;
50     for(int i=0;i<k;i++){
51         std::cout<<arr[i];
52
53         if(i<k-1){
54             std::cout<<" + ";
55         }
56
57         sum += arr[i];
58     }
59
60     Avg = (float)sum/k;
61     std::cout<<" = \""<<sum<<"\n"<<std::endl;
62     std::cout<<"Average Score:\n"<<sum<<" / "<<k<<" = \""<<Avg<<"\n"<<std::endl;
63     //genAvg
64     std::cout<<"=====REPORT====="<<std::endl;
65     std::cout<<"Higest Score: "<<H<<"\n"
66         <<"Lowest Score: "<<L<<"\n\n"
67         <<"Average Score: "<<Avg<<"%\n"
68         <<"Remark: "<<((Avg>74) ? "Passed" : "Failed");
69 }
70
71 int main() {
72
73     int n;
74     std::cout<<"Enter Class Size:"<<std::endl;
75     std::cin>>n;
76     std::cout<<std::endl;
77
78     int* dyna_arr = new int[n];
79
80     for(int i =0; i<n; i++){
81         dyna_arr[i] = input(i);
82     }
83
84     //while(
85     Report(n, dyna_arr);
86
87     delete[] dyna_arr;
88     return 0;
89 }
90

```

SOURCE CODE(TEXT):

```
/*
Developed by: DevEJ_Del_Rosario
Fullname: Edgardo B. Del Rosario Jr.
Class: BSIT-1C
=====
Project: 01 Laboratory Exercise - The ScoreKeeper (Dynamic Array)
Developed by: DevEJ_Del_Rosario
Copyright (c) 2026. All Rights Reserved.
=====
This code is the original work of Edgardo B. Del Rosario Jr.
Users may utilize this code as a guide or for educational purposes.
However, claiming this code as one's own or using it for commercial purposes
without proper citation or acknowledgment of the original creator is strictly prohibited.
=====

*/
#include <iostream>
int input(int n){
    int scores;

    // int arr[n];
    // Enter class size: 5;

    std::cout<<"Enter score #"<<n+1<<": ";
    std::cin>>scores;

    return scores;
}

void Report(int k, int arr[]){
    int sum=0;
    float Avg;

    // arr[k];
    int H = arr[0];
    int L = arr[0];

    for(int i=0;i<k;i++){

        if(H<arr[i]){
```

```

        H = arr[i];
    }
    if(L>arr[i]){
        L = arr[i];
    }
}

std::cout<<"\n\nTotal Score:"<<std::endl;
for(int i=0;i<k;i++){
    std::cout<<arr[i];

    if(i<k-1){
        std::cout<<" ";
    }

    sum += arr[i];
}

Avg = (float)sum/k;
std::cout<<" = \'"<<sum<<"\n"<<std::endl;
std::cout<<"Average Score:\n"<<sum<<" / "<<k<<" = \'"<<Avg<<"\n"<<std::endl;
//genAvg
std::cout<<"=====REPORT===== "<<std::endl;
std::cout<<"Higest Score: "<<H<<"\n"
    <<"Lowest Score: "<<L<<"\n\n"
    <<"Average Score: "<<Avg<<"%\n"
    <<"Remark: "<<((Avg>74) ? "Passed" : "Failed");
}

int main() {

    int n;
    std::cout<<"Enter Class Size:"<<std::endl;
    std::cin>>n;
    std::cout<<std::endl;

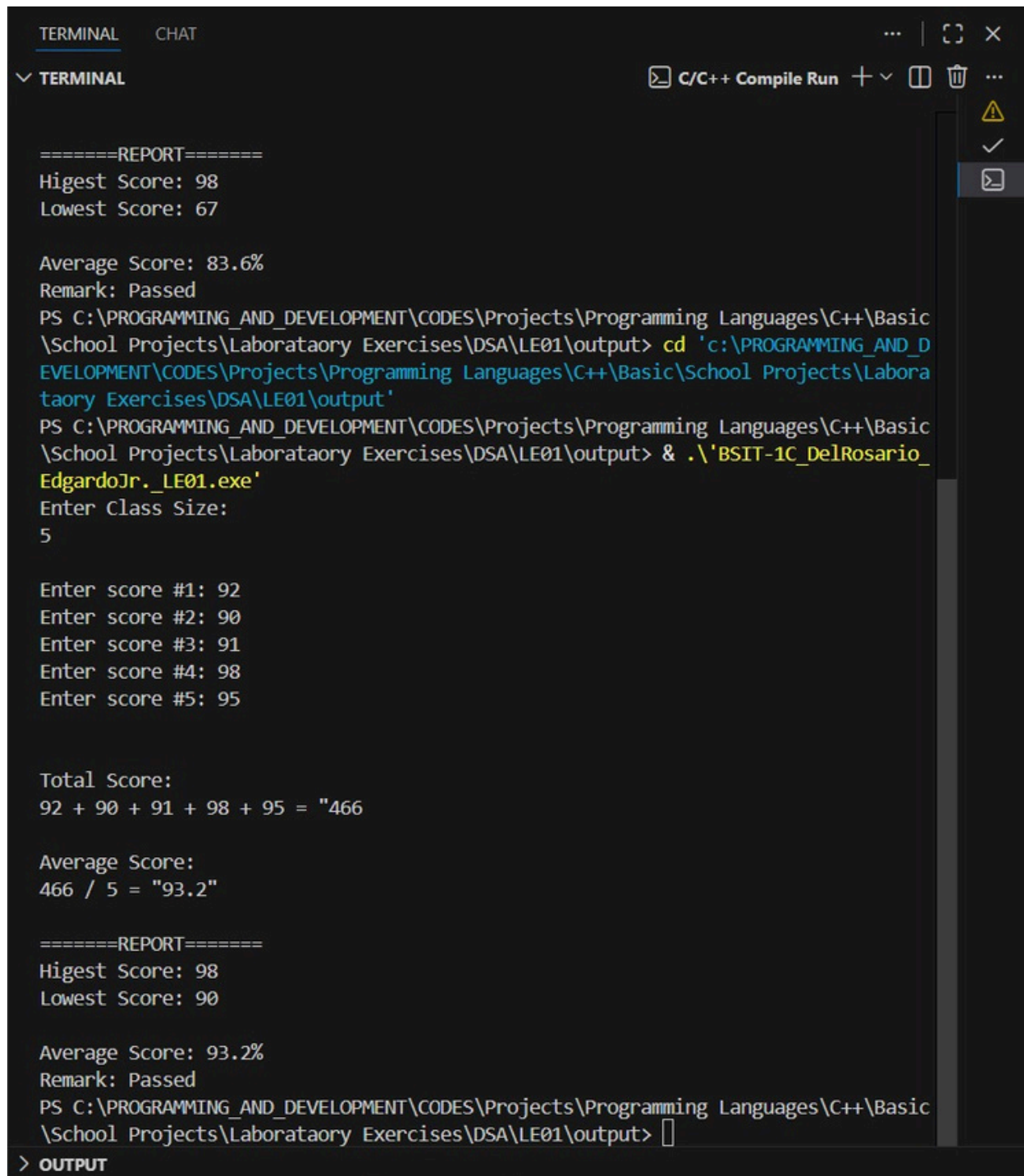
    int* dyna_arr = new int[n];

    for(int i =0; i<n; i++){
        dyna_arr[i] = input(i);
    }
    //while(
    Report(n, dyna_arr);

    delete[] dyna_arr;
    return 0;
}

```

PROGRAM SAMPLE OUTPUT:



The image shows a screenshot of a C/C++ IDE terminal window. The terminal has tabs for 'TERMINAL' and 'CHAT'. The 'TERMINAL' tab is active, showing the output of a C++ program. The program's output is as follows:

```
=====REPORT=====
Higest Score: 98
Lowest Score: 67

Average Score: 83.6%
Remark: Passed
PS C:\PROGRAMMING_AND_DEVELOPMENT\CODES\Projects\Programming Languages\C++\Basic
\School Projects\Laborataory Exercises\DSA\LE01\output> cd 'c:\PROGRAMMING_AND_D
EVELOPMENT\CODES\Projects\Programming Languages\C++\Basic\School Projects\Labora
taory Exercises\DSA\LE01\output'
PS C:\PROGRAMMING_AND_DEVELOPMENT\CODES\Projects\Programming Languages\C++\Basic
\School Projects\Laborataory Exercises\DSA\LE01\output> & .\'BSIT-1C_DelRosario_
EdgardoJr._LE01.exe'
Enter Class Size:
5

Enter score #1: 92
Enter score #2: 90
Enter score #3: 91
Enter score #4: 98
Enter score #5: 95

Total Score:
92 + 90 + 91 + 98 + 95 = "466"

Average Score:
466 / 5 = "93.2"

=====REPORT=====
Higest Score: 98
Lowest Score: 90

Average Score: 93.2%
Remark: Passed
PS C:\PROGRAMMING_AND_DEVELOPMENT\CODES\Projects\Programming Languages\C++\Basic
\School Projects\Laborataory Exercises\DSA\LE01\output> 
```

The terminal window also shows a 'C/C++ Compile Run' button and a 'TERMINAL' tab. The output is displayed in a dark-themed window with a light gray border.

